

AI Provenance and the Evaluation of Intellectual Work

Rational Risk, Social Meaning, and Human–AI Performance

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Abstract

AI can improve a piece of intellectual work while knowledge of its use can reduce the credit or trust that the work receives. Understanding that possibility requires separating what AI did to performance from what disclosure adds to evaluation. We connect these questions by asking what the evaluator intends to assess. Improvement relative to a producer’s unaided baseline may matter when the objective is better output; a different benchmark is needed when the objective is to measure unaided capability, assign intellectual credit, or allocate opportunities. A binary “AI was used” label supplies little information about those distinctions. Its relevance depends on the function AI performed, the authority the human retained, verification, and the object of evaluation. Negative reactions may reflect expected error or checking cost, an institutional requirement, social meanings concerning effort and authenticity, or a combination of these explanations. We examine how evidence can distinguish them without treating every penalty as stigma. We then follow evaluation back into production: anticipated disclosure can change whether and how people use AI, including whether they forgo useful assistance. As technology and institutions evolve, the practical question remains whether provenance-based treatment serves the evaluation’s stated purpose or substitutes a category judgment for more direct evidence.

Keywords: AI provenance; AI disclosure; human–AI performance; evaluative objective; benchmark problem; signaling; stigma

1. Introduction

Consider a researcher who uses AI to organize literature, explore counterarguments, and revise a manuscript. The researcher checks the claims, selects the arguments, and takes responsibility for the submitted work. A journal requires disclosure of that assistance. The manuscript may be better than the researcher could have produced alone, yet learning that AI was involved may lead an editor to trust it less or assign the author less intellectual credit.

The apparent conflict deserves examination. The editor may expect hidden errors or additional checking costs. The editor may also be judging the author’s contribution, rather than only the manuscript’s quality. Alternatively, the disclosure may suggest that the work required less effort and is therefore less deserving of recognition. These explanations call for different evidence and different responses. The fact that AI helped produce the work establishes none of them by itself.

We examine what knowledge of AI use should add to the evaluation of intellectual work. By *AI provenance*, we mean information about AI’s involvement in producing the work, including what it did and how its output was used. *Disclosure* is one way that information reaches an evaluator; provenance

may also be inferred or reported by someone else. The distinction matters because evaluators respond to what they know or believe about production, which may differ from the production process itself.

Two bodies of research help establish the problem. Human–AI performance studies examine how assistance changes the work. In a field experiment with professional consultants, AI improved speed and quality on tasks within the model’s capability frontier but reduced correctness on a task outside it [1]. A broader meta-analysis found gains relative to humans working alone, while human–AI combinations performed worse on average than the better standalone human or AI alternative in the studies examined [2]. Assistance can be useful without being the best available production method.

Disclosure studies ask a different question: how does knowledge of AI involvement change evaluation? Experiments report lower trust, legitimacy, perceived quality, or ethicality in some settings [3–5]. Journalism research finds a more varied pattern, including null and conditional effects [6]. Earlier studies of algorithm aversion and appreciation likewise show that machine involvement can affect reliance in either direction [7,8]. These findings make the evaluation response an empirical question rather than a necessary consequence of improved or impaired performance.

The two questions meet when an evaluator judges hybrid human–AI work. AI may change the artifact, the producer’s effective performance, and what the artifact reveals about the producer. Knowledge of AI use can then change judgments about any of those objects. An improved manuscript and a reduced assessment of unaided writing ability can coexist because they answer different questions. An improved, verified manuscript receiving less credit merely because assistance seems less worthy presents a different concern.

Our argument begins with the purpose of evaluation. That purpose determines the appropriate performance benchmark and the attributes for which provenance might supply evidence. We then distinguish expected technical risk, institutional requirements, and social meaning as explanations for provenance-based treatment. Stigma remains one possible concern, but its interpretation depends on whether adverse treatment follows evidence relevant to the stated objective or devalues work and people through a disfavored production label.

The relationship also extends beyond a judgment made after submission. If researchers expect disclosure to reduce recognition, they may avoid useful assistance, redirect it, or conceal its use. Evaluation can consequently change the work that reaches the evaluator. Sections 2–4 develop the performance comparison and the interpretation of provenance effects. Section 5 follows their consequences into production and institutional review. Sections 6 and 7 examine changing provenance value and the evidence needed to evaluate it. The conclusion returns to the researcher’s practical question: what should disclosure change, and on what grounds?

2. Performance, Provenance, and the Benchmark Problem

2.1 Performance and provenance are different questions

For the researcher, the first question is whether AI improved the manuscript. The comparison is with what the same researcher would have produced without assistance under comparable conditions. Let i identify the producer, and let $Q_i(a)$ denote performance on a specified task and measure, with larger values

indicating better performance. The assistance condition is $a = 1$ for a specified AI-assisted workflow and $a = 0$ for its unaided counterpart. Holding the task, available resources, and comparison period appropriately fixed, define the augmentation effect as

$$\Delta Q_i = Q_i(1) - Q_i(0). \quad (1)$$

A positive value means that the assisted workflow improves performance relative to this producer's unaided baseline. The quantity does not compare the result with an expert, a field standard, or the strongest available alternative. It defines a counterfactual comparison; observing one assisted manuscript alone cannot reveal what its author would otherwise have produced. An empirical estimate requires a design that supports that comparison.

The provenance question holds the completed work fixed instead. Let W denote a particular work, and let $E(W, d)$ be its evaluation under a specified objective, rating scale, and background information. Here $d = 1$ means that AI involvement is disclosed and $d = 0$ means that the disclosure is withheld. With higher ratings indicating more favorable evaluation, define

$$\Delta E(W) = E(W, 1) - E(W, 0). \quad (2)$$

This contrast asks what disclosure changes when evaluators receive the same work. Withholding disclosure leaves provenance unspecified; it does not establish that the work was produced without AI. A study comparing an AI label with an explicit human-only label therefore uses a different informational baseline. Studies must state which contrast they estimate.

The two quantities describe different changes. Assistance changes production and potentially performance; disclosure changes the information available to an evaluator. Their separation lets us ask whether improved work is rewarded, discounted, or judged on an attribute other than the performance measure used in the first comparison.

2.2 The benchmark follows from the evaluative objective

Before interpreting either change, we need to know what the evaluation is trying to accomplish. A professional analyst may be asked to produce the best reliable report with the tools available in practice. In that setting, the comparison concerns feasible production systems, including AI-assisted ones. A licensing examination may instead measure unaided capability. A hiring decision may rank candidates, while journal review may ask whether a manuscript meets a field standard and makes a useful contribution.

These objectives give the same work different evidentiary roles. A polished answer can be a valuable output and weak evidence of unaided writing ability. A manuscript can improve substantially relative to its author's baseline and still fail a publication threshold. Conversely, an assisted researcher can exceed that threshold even if a stronger expert could have produced better work without AI. The evaluator's objective determines which comparison answers the decision at hand.

The benchmark problem becomes visible when a comparison is carried from one purpose into another. Suppose an editor believes that an AI system performs worse than an expert researcher. That belief does not establish that using the system made this author's manuscript worse. It compares AI with the expert,

whereas the augmentation question compares the author’s assisted and unaided workflows. Whether the final manuscript merits publication remains a third question, answered against the journal’s standard.

Improvement relative to an individual baseline is therefore informative within its proper role. It supports a claim about augmentation, not an automatic entitlement to publication or credit. Likewise, institutional concern about unaided capability is relevant where that capability is the object of assessment. We can judge the use of a benchmark more clearly once its purpose is explicit.

2.3 Four performance–evaluation configurations

With the performance measure, benchmark, and disclosure contrast specified, assistance and evaluation can move in the same direction or in opposite directions. Table 1 organizes the four nonzero sign combinations. The performance and evaluation measures need not use the same units, and their signs alone do not establish whether treatment is warranted.

Table 1. Performance and provenance-evaluation configurations

Performance effect	Disclosure effect	Question raised by the combination
Positive	Positive	Does favorable treatment recognize relevant improvement, or add an AI halo?
Positive	Negative	Does the penalty reflect a different relevant attribute, or discount beneficial assistance?
Negative	Negative	Does adverse treatment track the actual deficiency and the evaluation’s objective?
Negative	Positive	Does favorable treatment reflect another relevant attribute, or unwarranted confidence in AI?

The positive-performance, negative-evaluation case brings the researcher’s concern into focus. Useful assistance may be penalized. But an output gain could also accompany a reduction in what the work reveals about unaided capability, or a breach of an assistance rule. To distinguish these explanations, we must follow the evaluation to its intended object rather than infer bias from the sign combination.

The opposite case provides a corresponding warning. A recipient may associate AI with sophistication or analytical power and favor work whose measured performance has worsened. Such an effect would be problematic if the decision is meant to reward that performance. Positive provenance meaning deserves the same evidentiary scrutiny as negative meaning.

Benchmark selection and performance assessment are thus separate tasks. First establish the objective and comparison; then measure what assistance changed. Only after those steps can we ask what the disclosure response adds. This leads to a more specific question about the label itself: which attributes does “AI was used” allow an evaluator to infer?

3. What AI Provenance Reveals

3.1 Provenance is a cue, not the attribute itself

A binary AI label supplies little detail about the production process. The researcher in our example may have used AI to organize sources and challenge an argument while retaining substantive control. Another author may have accepted generated claims and citations with little scrutiny. Both can truthfully report that AI was used, although the resulting error risks and human contributions may differ substantially.

Evaluators fill these informational gaps with inferences. They may expect more errors, a larger checking burden, weaker oversight, less original contribution, lower effort, or reduced authenticity [3,4,5,9]. We call provenance a *cue* because it prompts such judgments without directly measuring the attribute at issue. Its usefulness depends on how well the inference fits the actual use and the evaluation's purpose.

This distinction explains why a single attitude toward AI cannot adequately describe provenance effects. An editor can accept AI-assisted language revision while questioning unverified generated references. A teacher can value the same revision tool in practice but exclude it from an exercise designed to assess unaided writing. The label remains the same; its relationship to the attribute being assessed changes. The journalism review likewise relates provenance cues to the inferences audiences draw and the objects they judge [6]. We next apply that distinction to the evaluation of intellectual work.

3.2 What is being evaluated?

We can make that relationship clearer by identifying the target of evaluation. For the artifact, relevant attributes may include accuracy, reasoning, originality, and usefulness. For the producer, an evaluator may assess competence, intellectual contribution, independent reasoning, or trustworthiness. For the process, compliance, transparency, reproducibility, and retained human authority may matter. A separate judgment can concern the institution's own use and governance of AI.

These targets often coexist. An editor may judge manuscript quality and assign authorial credit in the same decision. A student answer may be correct but offer insufficient evidence of independent reasoning. A candidate's résumé may accurately communicate qualifications while providing little evidence of unaided writing skill. A single overall rating can conceal those differences.

The practical response is to identify the attribute before interpreting the provenance effect. If a technically sound manuscript is rated less favorably, did perceived accuracy decline, did confidence in the author's competence decline, or was less credit assigned for the contribution? Each response could require a different explanation. The artifact's signaling role helps explain why judgments about producer capability can change even when the artifact itself remains unchanged.

3.3 Signaling and credentialing

A work product can have value in itself and reveal something about the person who produced it. An institution may use a writing sample, examination answer, or research paper to infer a capability that is otherwise difficult to observe. Signaling theory provides a precedent for understanding how observable achievements convey information about less directly observable characteristics [10].

AI can change that relationship. If assistance raises the performance of less capable unaided producers more than that of stronger producers, their visible work may become more similar. This is a conditional possibility, not a universal description of AI use. Where it occurs, an institution obtains better outputs but less separation among underlying unaided capabilities.

Whether that change is desirable depends on the objective established in Section 2. A buyer seeking a reliable report may welcome broader access to strong production. An examiner seeking to distinguish unaided capability needs a different assessment. The improved report does not become less useful merely because it is less diagnostic of unaided skill, and the examiner's concern does not disappear merely because the answer improves.

The signaling explanation gives provenance a particular function: it helps interpret the mapping from observed work to the human attribute being inferred. That function is strongest when the assessment truly concerns unaided capability. If the work will normally be performed with AI, competence in using, checking, and directing AI may itself belong in the assessment. Institutions need to specify which capability they wish to credential rather than leave the choice implicit.

3.4 Function and retained authority

Knowing what AI did makes the cue more informative, but function alone still leaves a central question unresolved: who exercised judgment over the result? Language correction, coding, retrieval, synthesis, hypothesis generation, substantive drafting, and autonomous decisions involve different activities. Within each activity, people can retain or relinquish different degrees of authority.

For example, substantial generated text can remain subject to human source selection, rejection, recombination, and verification. A small generated passage can carry considerable influence if the author accepts its reasoning without scrutiny. The amount of generated text consequently provides an incomplete measure of either contribution or accountability. We need to know what the human understood, adopted, checked, and took responsibility for.

A useful disclosure connects those details to the evaluation. Verified references address citation reliability. A record of source selection and argument development helps explain contribution. A statement that AI was limited to language revision may inform a research-quality judgment while still identifying prohibited assistance in an unaided writing assessment. Specificity has value because it resolves a relevant uncertainty, not simply because it makes the disclosure longer.

Specific disclosure may also reveal which explanation remains unresolved. Credible verification can reduce concern about factual error while leaving judgments about effort or authenticity unchanged. Evidence of substantive human control may address contribution without satisfying a human-only production requirement. Those remaining differences prepare the next inquiry: what mechanism explains a provenance penalty, and what evidence would distinguish it from alternatives?

The same reasoning applies to identical-output experiments. Holding the artifact fixed isolates an effect of provenance information on evaluation. It does not hold constant every inferred attribute of the producer or process. Such studies provide strong evidence that the cue affects judgment; whether the effect is

inappropriate depends on the attribute the evaluator is meant to assess and what information the cue legitimately supplies about it.

4. From Competing Explanations to Residual Provenance Effects

4.1 Three classes of explanation

We begin with three possible explanations for a provenance response. A statistical explanation uses AI involvement to predict an attribute such as error risk or checking cost. An institutional explanation follows from the assessment’s purpose or assistance rules. A social-meaning explanation concerns the effort, authenticity, legitimacy, or deservingness associated with AI use. Here *legitimacy* means perceived appropriateness within the relevant norms, and *authenticity* concerns whether the work expresses the human origin or agency valued in that setting.

The explanations can overlap. An authenticity judgment may be central to a human-only competition and peripheral to the accuracy of a forecast. A contribution judgment can reflect both an authorship requirement and a social belief that assisted work is less worthy. The question is how a particular judgment connects to the stated objective and the evidence supporting it.

Table 2 identifies comparisons that can help distinguish these explanations. The comparisons are diagnostic starting points, not a procedure for subtracting a “rational” amount from a total penalty.

Table 2. Explanations and diagnostic comparisons for provenance effects

Explanation	Illustrative concern	Diagnostic comparison
Statistical	AI use predicts unreliable citations or additional checking cost	Compare the response when credible verification or direct burden evidence is supplied; test whether provenance still predicts the relevant risk.
Institutional	An examination measures unaided reasoning, or a task restricts assistance	Compare assessments with different stated purposes or assistance rules while preserving relevant task conditions.
Social meaning	AI involvement changes perceived authenticity, commitment, or deservingness	Test those judgments alongside performance, contribution, and compliance evidence; determine whether they explain adverse treatment beyond the stated evaluative requirements.

4.2 Artifact risk and verification burden

For an editor assessing a manuscript, an error-risk explanation is straightforward. If a type of AI-assisted workflow more often produces fabricated references or unsupported claims, knowledge of that workflow can alter the editor’s expectation of quality. The relevant empirical object is the relationship between the actual workflow and the failure, rather than the mere presence of AI.

Checking cost gives a related but distinct explanation. Two manuscripts could ultimately prove equally accurate while requiring different amounts of verification. Under limited review capacity, expected

checking burden can affect review decisions even when final accuracy is the same. A study that controls accuracy alone may therefore leave a decision-relevant cost uncontrolled.

The binary label is a weak substitute for those details. As credible direct evidence becomes available, the case for relying on a generic label should weaken to the extent that the evidence resolves the concern. Validated references are more informative about those references than a broad assumption about AI-assisted writing. They do not automatically establish the reliability of every other claim in the manuscript. The appropriate shift from proxy to evidence follows the scope of what was actually checked.

4.3 Contribution, accountability, and compliance

Verification addresses the artifact but may leave the author's role unresolved. If authorship entails understanding, adopting, and accepting responsibility for claims, the editor has reason to ask how the human exercised those functions. A strong artifact alone may not establish who supplied the substantive judgment or can defend it.

Compliance introduces another question. An assisted answer may be accurate while violating a clear rule about permitted tools. That rule supplies a reason for adverse treatment independent of artifact quality. Whether the rule serves a sound objective remains open to examination; a rule's existence and its justification are distinct issues.

During technological transition, incomplete or uneven guidance makes both questions harder. A lifecycle-based study of health-related journals found that 72.6 percent prohibited AI authorship while 54.2 percent supplied no guidance on methodological design [11]. These findings illustrate uneven coverage of AI's roles across the research process. They do not show that every provenance penalty follows policy or that every policy is justified.

An institutional explanation should identify the requirement involved: unaided capability, accountable authorship, confidentiality, or another specified obligation. A general appeal to "journal policy" may explain why a rule was applied without explaining why that rule belongs in the evaluation. Making the requirement explicit allows the policy itself to be assessed as practices and available evidence change.

4.4 Effort, legitimacy, and social meaning

Once risk, contribution, and compliance are clearer, an evaluator may still assign less value to assisted work because it appears easier to produce. The role of effort needs examination. Effort can be part of an educational exercise, a proxy for care and commitment, or a symbolic source of worth. These roles lead to different judgments about the same reduction in labor.

In an outcome-oriented analysis, reducing effort while preserving or improving quality can represent productivity. In an exercise intended to develop a skill, bypassing the relevant practice can defeat the purpose. If effort is used as evidence of care, credible verification and sustained human judgment may be more informative than the number of hours spent drafting. The evaluative objective determines which relationship matters.

Research on the effort heuristic provides a precedent for greater inferred effort raising perceived value [9]. AI-disclosure studies connect reduced trust with judgments about legitimacy, commitment, and

authenticity [3,4,12]. These findings help explain why technical assurances may leave some provenance effects intact. They also direct attention to the question motivating this paper: is assisted intellectual work being discounted for a relevant deficiency, or because the manner of production has acquired a disfavored social meaning?

4.5 Progressive control and the interpretation of stigma

We can investigate that question by progressively testing the most plausible explanations. For the researcher's manuscript, begin with the editor's stated concern. If it is reference reliability, provide credible verification and examine whether the response changes. If it is contribution, clarify the human role. If it is compliance, specify the applicable rule and whether the use met it. Measure performance against the appropriate benchmark rather than assume that polished prose establishes quality.

A smaller penalty after such information is supplied supports the view that the addressed uncertainty contributed to the response. A persistent penalty creates a further question; it does not, by itself, identify stigma. Evaluators may distrust the assurance, care about another relevant attribute, or apply a requirement the study has not adequately represented. Research must establish what information was credible and what the assessment was meant to determine.

We use *stigma* here for social devaluation attached to AI involvement that extends beyond a demonstrated deficiency relevant to that assessment. The interpretation becomes stronger when credible evidence resolves performance, contribution, and compliance concerns, yet adverse treatment remains connected to judgments that assisted work or its producer is less worthy. Evidence about the proposed social mechanism is needed alongside evidence that alternative explanations have weakened.

This is a progressive evidentiary approach rather than a numerical decomposition. Risk beliefs, institutional requirements, and social meanings can influence one another. Measured mediation is informative but does not alone establish a causal mechanism; carefully designed manipulations and comparisons can provide stronger tests. There is no uniquely identified "stigma coefficient" merely because a regression leaves a residual provenance effect.

The institutional implication follows from the same reasoning. When direct evidence is costly, provenance may serve as an inexpensive proxy. As better measures become feasible, institutions can compare the proxy with the attributes it is intended to predict. Reference validation, process documentation, oral defense, and observed review burden address different uncertainties. Retaining an old proxy after it loses value may reflect inertia or switching costs, but its continuing use still requires a connection to the decision.

We should apply the same test to favorable provenance meaning. Recipients may infer greater sophistication or accuracy from AI involvement even when the work does not support that inference. Algorithm appreciation provides a precedent for greater reliance on algorithmic advice [8]. An evaluation system should examine both undeserved devaluation and undeserved confidence, while preserving legitimate judgments about the work, person, or process.

So far, we have considered a completed artifact and the information supplied about it. The researcher, however, may anticipate these judgments before beginning the manuscript. That expectation makes evaluation part of the production problem.

5. Provenance Regimes Can Change Production

5.1 Anticipated disclosure changes the production pathway

If a researcher expects disclosure to reduce recognition, the expectation can change how the manuscript is produced. The researcher may avoid AI, limit it to accepted functions, verify more carefully, conceal its involvement, or leave more of the task to the system. Different responses can improve or impair the result. The direction depends on how the producer interprets the expected treatment.

Evidence for such an indirect effect comes from experiments involving text-to-image creative collaboration [13]. Creators anticipating disclosure often withdrew from the creative process, and the resulting artifacts reflected less human creative agency. This finding establishes a production-stage response in that setting. Whether researchers, students, or professional analysts respond in the same way requires further evidence.

The general mechanism is that the disclosure regime supplies an incentive before the artifact exists. We call this an *endogenous production–evaluation relationship*: evaluation expectations arise within, and can influence, the process that produces the work later evaluated. Figure 1 distinguishes the artifact pathway from the provenance-information pathway and shows how observed treatment can inform future expectations.

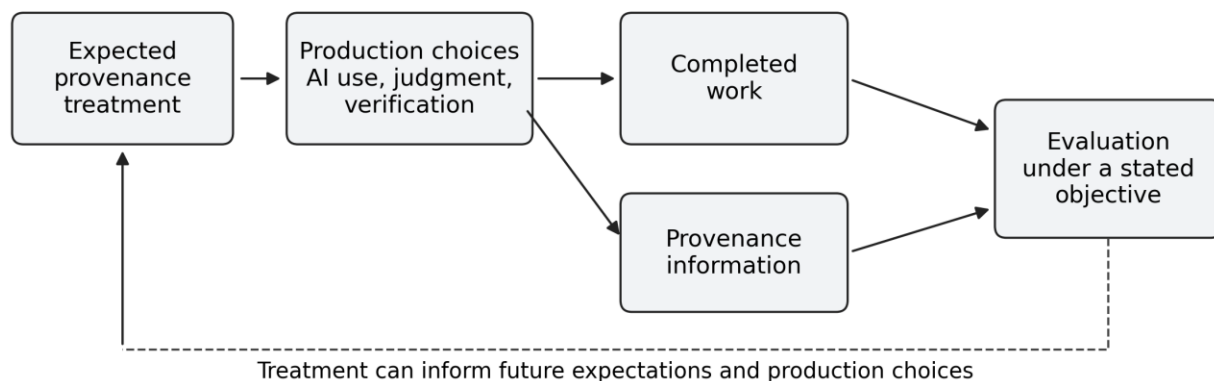


Figure 1. Anticipated provenance treatment can change production before evaluation.

The diagram represents possible pathways rather than measured effects. A disclosure label can influence evaluation directly, while anticipated disclosure can affect the artifact through production choices. Past evaluation can then alter expectations for later work. This is why comparing disclosed and undisclosed manuscripts in routine submissions may combine a response to the label with differences already created during production.

5.2 Producer responses and useful assistance forgone

The researcher now faces a choice about both production and disclosure. Using and disclosing, using and concealing, avoiding AI, and redirecting assistance toward accepted functions carry different performance and reputational consequences. Educational evidence links fear of judgment with concealment [14].

Research on the disclosure paradox also shows that support for transparency can coexist with unfavorable reactions to disclosed use [12]. A producer can consequently perceive a cost to complying even when the institution favors disclosure.

One concern is useful assistance forgone. Suppose AI would improve the accuracy or reasoning of a manuscript, human contribution remains sufficient, and the use satisfies the relevant rules. If expected devaluation deters assistance, the institution may receive worse work because of an incentive generated by its evaluation practices. The loss is consequential precisely because assistance would have served the institution's stated objective.

The same reasoning identifies a different case. If an examination is designed to measure unaided reasoning, foregoing AI preserves the assessment rather than sacrificing its objective. We need the objective and feasible workflow to distinguish discouraged improvement from appropriate restraint. Expected disclosure costs become problematic when they work against the outcome the institution intends to obtain.

Concealment creates another consequence. Institutions may infer the risk of AI-assisted work from disclosed cases even though some assistance remains hidden. Once disclosure depends on expected treatment, the observed group is selected. The next question is how institutions use that group to allocate scarce evaluation resources and update their beliefs.

5.3 Institutional triage and selected evidence

Lower production costs may increase the supply of manuscripts, applications, and proposals faster than expert review capacity. Under such conditions, an institution may seek inexpensive cues for deciding what to check, route, prioritize, or reject. AI status is a possible cue if the institution believes it predicts error, checking burden, or compliance risk. Whether it actually serves that function requires operational evidence.

The distinction is between using provenance for a demonstrated relationship and using an easily observed category in place of that relationship. If AI status predicts additional checking cost after relevant manuscript characteristics are considered, it may help allocate review effort. If direct measures account for that cost and the label adds little, a category rule may be a poor substitute. Merely collecting disclosure does not save review resources; the information must change the review path to have an operational effect.

Producer responses can distort the evidence behind the rule. Concealment may remove some AI-assisted work from the disclosed group. Avoidance may remove other producers from AI use entirely. Disclosed cases might overrepresent highly compliant authors, settings where disclosure is unavoidable, or workflows requiring extensive assistance. These possibilities imply different observed associations

between disclosure and quality. Institutions cannot infer the causal effect of assistance from such associations without addressing selection.

A feedback process can then develop. An institution uses disclosure to guide review; producers respond to the treatment; those responses change the submitted and disclosed populations; the institution learns from the altered sample. The resulting beliefs may reinforce or weaken the original rule. Testing the rule therefore requires evidence about production and disclosure behavior as well as the review outcomes of visible cases.

An institution can pursue its objective coherently under resource constraints while still using a contestable objective or an imperfect proxy. Here, *institutional rationality* refers to that coherence, not to moral neutrality or freedom from bias. The relevant questions remain whether the objective is justified and whether the provenance rule serves it under current conditions.

Those questions become especially visible when AI operates on both sides of evaluation. A candidate may use AI to revise a résumé that an employer screens with AI; an author may disclose assistance to a journal using automated triage. These are illustrative arrangements, not evidence that the functions are equivalent. Different roles can justify different rules, but the justification should identify those roles and risks. The presence of the same technology alone explains neither consistency nor inconsistency.

6. The Evolution of Provenance Value

The feedback just described also makes provenance a changing signal. Its value depends on the reliability of models, the way people use them, available verification, institutional rules, and the meanings attached to assistance. Changes in one condition need not produce changes in the others.

For example, improved reliability can weaken an error-risk inference while leaving a requirement for unaided work intact. Clearer rules can reduce compliance uncertainty while arguments about deservingness continue. Wider access can reduce one source of competitive disadvantage without improving every user's judgment. An observed overall penalty can stay stable while its underlying explanations change.

The dynamic proposition is consequently about the composition of the cue's value. When the conditions supporting a provenance inference change, the relevance of that inference should be reassessed. Current disclosure effects describe particular technologies, uses, audiences, and institutions rather than a permanent property of all AI-assisted work.

6.1 A limiting case: widespread adoption

Consider a hypothetical future in which 90 percent of serious intellectual work materially depends on AI assistance. The figure is an illustrative assumption, not an adoption forecast. A generic presence label would identify a large majority of work and supply little detail for distinguishing uses within it. Treating all assisted work as an exceptional review category would also burden institutions seeking to evaluate ordinary production.

The resulting need is for finer information: what AI did, who retained authority, what was verified, and which risks remain relevant. Widespread use does not by itself eliminate differences between assisted and unaided work. Where unaided capability or human-only production is the objective, provenance still distinguishes a relevant condition. Indeed, human-only production could become the salient exception.

The limiting case therefore exposes the weakness of presence as a universal evaluation rule. Prevalence changes how much the label sorts the population, while purpose determines whether that sorting matters. Neither widespread adoption nor improved average reliability supplies one answer for examinations, manuscript review, professional analysis, and authenticity-sensitive work.

6.2 From technology presence to evaluation design

As use becomes routine, institutions can ask for information closer to their decision. A research review may need verification of substantive claims and an account of human responsibility. A skills assessment may need controlled unaided performance. An operational forecast may need validation of the complete assisted workflow. These are different evaluation designs, each answering a question that a binary label leaves unresolved.

Normalization can also change social meaning without settling technical concerns. AI assistance may become unremarkable in one setting while human origin acquires greater value in another. Existing disclosure research already shows context-dependent responses and legitimacy concerns [3,4,6]. Normalization can therefore produce different trajectories across settings rather than a uniform movement toward acceptance.

Evaluation capacity supplies a further reason to track several conditions separately. A lower error rate per use can coexist with more total errors if use expands sufficiently. More capable users can coexist with congested review systems. Better governance can coexist with disagreement about intellectual credit. Studies of change should measure the relevant attributes and mechanisms rather than treat time as a single measure of technological maturity.

These considerations lead to a research program that follows performance, provenance information, producer responses, and institutional decisions together. Measuring only whether the average disclosure penalty grows or shrinks would leave the principal explanations unresolved.

7. Implications and Research Priorities

7.1 Specify the evaluative objective and target

The first requirement is to state what the decision is intended to assess. Artifact quality, unaided capability, intellectual contribution, process compliance, and institutional trust can respond differently to the same information. A study should identify the target and record those judgments separately where they perform different roles.

The performance benchmark follows from that choice. Researchers should specify whether they compare a producer with an unaided counterpart, rank candidates, apply a common field standard, or compare

feasible production systems. This makes apparently conflicting findings interpretable and prevents a result about augmentation from being used as a conclusion about credentialing or publication quality.

7.2 Replace binary presence with functional information

Having identified the decision, researchers can ask which production details address its uncertainty. Disclosure conditions should distinguish relevant functions, retained human judgment, verification, and compliance. The goal is to learn which information changes the evaluation and why.

For citation reliability, compare an AI label with credible evidence about the references. For contribution, compare the label with a documented account of source selection, reasoning, and human review. Direct evidence deserves greater weight to the extent that it validly measures the attribute; a producer's unverified assurance is itself another cue. Designs should make the credibility and scope of the information explicit.

7.3 Measure performance and provenance together

A central empirical opportunity is to measure assistance and evaluation in the same study. Randomly assigning AI access on a specified task can estimate the effect of access on performance. Actual use should also be recorded: access does not guarantee use, and comparing users with nonusers after assignment does not automatically preserve the causal interpretation.

After production, independent evaluators can receive the same artifact with or without truthful information about AI involvement. Randomizing that information holds the artifact fixed and estimates the disclosure contrast. Separate conditions can provide credible verification, contribution, or compliance information. The design can then examine whether evaluation responds to actual performance, other relevant attributes, or social meanings attached to provenance.

It is important to retain the distinction between access and disclosure. AI access changes a production opportunity; disclosure changes evaluator information. Their effects can be estimated separately under the relevant randomizations, while their relationship can be compared across tasks and objectives. Both negative penalties and positive halos belong in that comparison.

7.4 Measure expectations, behavior, and operational treatment

A post-production label experiment leaves anticipated disclosure untested. To study the pathway in Section 5, researchers must vary disclosure expectations before work begins and measure the resulting choices. Relevant outcomes include adoption, prompting, verification effort, concealment, task choice, and submission. The creative-collaboration evidence provides a precedent for this production-stage design [13], not a substitute for testing intellectual-work settings.

Institutional studies can examine what disclosure changes in practice: desk rejection, review priority, verification requests, processing time, and acceptance. Recorded quality and burden measures can help assess the usefulness of the provenance flag. Observational adjustments should be interpreted in light of selection into use, disclosure, and submission; controlling a few observed characteristics does not establish the absence of those differences.

7.5 Replicate as conditions change

The same study should be repeated as model reliability, AI literacy, use patterns, rules, and review infrastructure change. These conditions supply possible explanations for changing provenance effects. Repeated designs should retain comparable objectives and measures while recording the conditions that have evolved.

A stable overall rating effect may conceal less concern about error and more concern about authenticity. A declining effect may reflect better verification rather than greater approval of AI use. Measuring the separate judgments allows researchers to distinguish these interpretations and gives institutions evidence for revising particular rules.

7.6 Integrating the research program

A manuscript-production study illustrates how the priorities fit together. Producers could be assigned AI access or no access and, independently, a specified expectation about later disclosure. The task, performance standard, and permitted assistance would be made explicit. Researchers would record actual tool use and verification, then assess the resulting work using measures suited to the stated objective.

Only after the work is complete would evaluator information be randomized. For assisted manuscripts, evaluators could receive either truthful disclosure or no provenance information, with further conditions describing verified claims and the human role. This separates a label response from production changes associated with prior expectations. An explicit human-only label would constitute another contrast and should be analyzed accordingly.

The evaluation could separately measure manuscript accuracy, contribution credit, inferred author capability, checking burden, and the review decision. That separation would reveal whether disclosure affects the artifact judgment, the producer judgment, or the review process. It would also show whether improved output and reduced credit coexist for reasons connected to the assessment's purpose.

Operational journal or employer data could complement the experiment by showing how such judgments translate into routing and resource allocation. The strongest interpretation would combine credible performance comparisons, clear evaluative objectives, evidence about the actual human role, and tests of competing explanations. A remaining adverse response might support a stigma interpretation, relevant caution, mixed mechanisms, or no consequential effect. The research question is which account the evidence supports.

7.7 Scope conditions

Our analysis concerns intellectual work where provenance can influence evaluation and evaluators retain discretion. It organizes comparisons and competing explanations rather than supplying a universal evaluation rule. The broad category "AI" includes systems and workflows whose differences must be specified in empirical applications. Safety-critical deployment requires additional task-specific analysis.

The literature and policy evidence retained here reflects the manuscript's evidence base through September 3, 2026. The changing conditions discussed in Section 6 are reasons to revisit that evidence, not claims that later developments have already confirmed the proposed pathways. The empirical studies

cited illustrate particular effects; the integrated research program remains a proposal for testing their connections.

8. Conclusion

The researcher's disclosure raises a question about both evidence and intellectual credit: what should knowledge of assistance change? A sound answer begins with what the evaluator intends to determine. Better output, unaided capability, accountable contribution, and compliance are different objects, even when one institution considers several in the same decision.

AI may improve work relative to the producer's unaided baseline without making the result the best available work or sufficient for publication. Equally, the fact that AI performs worse than an expert does not establish that it worsened a particular author's manuscript. The benchmark must fit the claim being made. Once that comparison is clear, we can examine whether provenance supplies additional evidence about the relevant attribute.

Error risk and checking cost can justify scrutiny where the actual workflow predicts them. Authorship and assistance rules can make production details directly relevant. Social meanings concerning effort, authenticity, and deservingness can also influence treatment. Those meanings deserve examination when they discount assisted work or its producer beyond the deficiencies relevant to the assessment. Credible direct evidence and tests of alternative mechanisms strengthen that inquiry; a penalty alone does not settle it.

Evaluation practices also create incentives. A provenance rule can improve verification, preserve an unaided assessment, discourage useful assistance, or alter what producers disclose. Institutions then learn from a population partly shaped by their own rules. Understanding that feedback is necessary for judging whether the rule serves its intended purpose.

As intellectual production becomes more hybrid, the useful question is increasingly specific: what did AI do, what judgment did the human retain, what was checked, and how do those facts bear on this decision? Asking those questions preserves legitimate scrutiny while making evaluation more responsive to the quality, contribution, and responsibility the work actually embodies.

AI-Use Disclosure

Generative AI tools were used during the development of this work to support literature discovery, conceptual exploration, drafting, editing, and review. The research questions, analytical framework, interpretation of evidence, selection of arguments, and final judgments remain the responsibility of the author. References and substantive claims were checked against source materials where applicable.

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